Phase 1:Brainstorming & Ideation

SmartLender: AI-Powered Credit Card Approval System

Role	Name
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Project Title

SmartLender: AI-Powered Credit Card Approval System

Problem Statement

Banks and financial institutions receive thousands of credit card applications every day. A large share of these are rejected because of factors such as high existing loan balances, insufficient income, or excessive credit inquiries. Reviewing every application by hand is slow and prone to human error, which makes it difficult for a bank to scale its approval process while keeping decisions consistent across applicants.

Proposed Solution

SmartLender automates the credit card approval decision using machine learning. A predictive model is trained on historical applicant data covering income, employment, and credit history, a evaluates new applications in a manner comparable to a human credit analyst. Four classification algorithms are compared – Logistic Regression, Random Forest, XGBoost, and Decision Tree – and the strongest performer is deployed through a Flask web application, with an optional IBM Watson Machine Learning pipeline available for cloud hosting.

Target Audience

- **Primary:** Bank credit analysts and compliance officers who need quick, reliable decision support.
- **Secondary:** Prospective customers who want to check their eligibility before formally applying.

Technology Stack

- Python, NumPy, Pandas – data handling
- Scikit-Learn, XGBoost – model training
- Matplotlib, Seaborn – exploratory analysis
- Flask – web application
- IBM Watson Machine Learning – optional cloud deployment

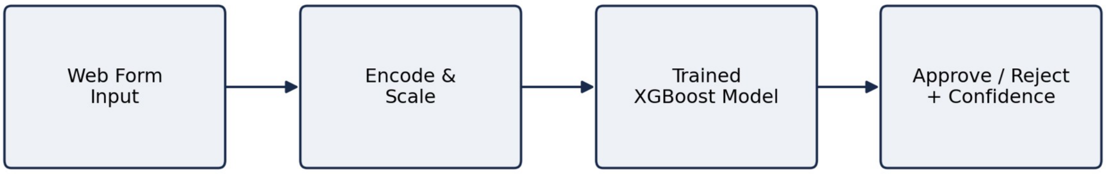

Figure 1: High-level prediction flow